



Curriculum Vitae

In So Kweon

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Biography

Professor In So Kweon received the B.S. and the M.S. degrees in Mechanical Design and Production Engineering from Seoul National University, Korea, in 1981 and 1983, respectively, and the Ph.D. degree in Robotics from the Robotics Institute at Carnegie Mellon University in 1990. He worked for Toshiba R&D Center for one year and joined KAIST in 1992. He is now a KAIST Professor at the School of Electrical Engineering (EE) at KAIST. He was a KEPCO Chair Professor of the EE and had been the director for the National Core Research Center – P3 DigiCar Center at KAIST (2010~2017). His research focuses on Computer Vision and Robotics. He has published 3 research books, and more than 500 papers in leading journals and conference proceedings, including 100+ in prestigious CVPR, ICCV, and ECCV. He was awarded “2016 Faculty Research Excellence Award”, “2020 Grand Prize for Academic Excellence”, and “2021 Hyung-Gyu Im LINKGENESIS Best Teacher Award” by KAIST and was conferred a Prime Minister Award by the Korean Government for his contribution to DRC-HUBO+ to win the DARPA Robotics Challenge in 2015. He also received several awards from international conferences, including "The Best Student Paper Award of the IEEE/CVF WACV 2023, “The Best Paper Award of the IEEE Transaction on CSVT 2014" and "The Best Student Paper Runnerup Award in the IEEE-CVPR 2009".

Academic Activities

He has also been active in professional services. He was the President of the Asia Federation of Computer Vision (AFCV) from 2016 to 2022. He served on the Editorial Board of the International Journal of Computer Vision for ten years since 2005. He has also organized 5 international conferences as a general or program chair, including IEEE/CVF ICCV 2019.

RESEARCH INTERESTS

Computer Vision and Robotics
(Deep Learning, Object Recognition, 3D reconstruction, Stereo systems, Robust features)

EDUCATION

- Aug. 1990 Ph.D. in Robotics, The Robotics Institute, Carnegie Mellon University
 - Advisor: Prof. Takeo Kanade
 - Thesis title: Modeling Rugged Terrain by Mobile Robots with Multiple Sensors
- Feb. 1983 M.S., Dept. of Mechanical Design and Production Engineering, Seoul National University
 - Advisor: Prof. Dong Won Kim
- Feb. 1981 B.S., Dept. of Mechanical Design and Production Engineering, Seoul National University

WORK EXPERIENCE

- 1992~present Professor, School of Electrical Engineering, KAIST
- 2017~2018 Visiting professor, Samsung Research, Seoul
- 2015~2018 KEPCO Chair Professor, School of Electrical Engineering, KAIST
- 1998~1999 Visiting professor, Department of Engineering, Cambridge University
- 1991~1992 Researcher, Toshiba R&D Center, Japan.
- 1983~1984 Researcher, NC Center, KIMM, Korea.

AWARDS & HONORS

Grand Academic Excellence Award (학술대상), 50th Anniversary, KAIST, 2021.

Special Award (최다논문제출 지도교수), Samsung Humantech Award, 2019.

Best Paper Award, Domestic Journal of Korea Robotics Society (KROS), 2018.

Winner, Chalearn LAP Inpainting Competition Track 2 - Video decaptioning, European Conference on Computer Vision (ECCV), 2018.

4D Light-field Depth Challenge: Robustness Champion, IEEE Computer Vision and Pattern Recognition (CVPR), 2017.

Best Poster Award, 29th Image Processing and Image Understanding (IPIU) Workshop, 2017

The Technology Innovation Award of KAIST, Korea, December, 2015.

Best Poster Award, International Workshop on Robust Computer Vision, Asian Federation of Computer Vision, Beijing, China, November, 2015.

Winner (as PI of the vision system of DRC-HUBO+ by Team KAIST), The DARPA Robotics Challenge Finals, USA, June, 2015.

Best Paper Award, IEEE T. Circuits and Systems for Video Technology (CSVT), 2014

Best Poster Award, Workshop on Robots, Korea Robotics Society, Korea, 2014

Best Poster Award, Image Processing Image Understanding Workshop, Jeju, Korea, 2014

Best Poster Award, International Workshop on Robust Computer Vision, Asian Federation of Computer Vision (AFCV), Daejeon, Korea, 2013.

Best Student Paper-Runner Up Award, IEEE Computer Vision and Pattern Recognition (CVPR), Miami, USA, 2009.

Student Paper Award, International Conference on Control, Automation and Systems (ICCAS), Seoul, Korea, 2008.

Outstanding Poster Award, International Conference on Ubiquitous Robots and Ambient Intelligence (URAI), Seoul, Korea, 2008.

Best Poster Award, Workshop on Robots, Korea Robotics Society, Jeju, Korea, 2007

Poster Paper Award, SPIE Photonics Conference, Boston, USA, 2002

KAIST Research Award, Korea, 2001

Invited Talks

Sep 2021	Plenary speech, Global Summit and EXPO on Robot Intelligence Technology and Applications (GSERITA), Lisbon (Virtual), 2021
Jun 2021	Keynote speech, Fifth International Workshop on Symbolic Neural Reasoning, Osaka (Virtual), 2021
Dec 2017	Keynote speech, IEEE Conference on Visual Communications and Image Processing, 2017
Oct 2016	Keynote talk, IEEE Conference on Robots and Systems (IROS), 2016
Nov 2015	Keynote speech, Annual Workshop on Korea Multi-media Society, Daegu, Korea, 2015
Oct 2015	Keynote speech, GPU Technology Conference, Seoul, Korea, 2015
Sep 2015	Keynote speech, Korean Conference on Computer Vision, Seoul, Korea, 2015
Sep 2013	Keynote speech, Internal Workshop on DSP-in-Vehicles 2013
Oct 2013	Invited talk, Asian Conference on Pattern Recognition (ACPR) 2013
Aug 2012	Plenary talk, SICE Annual Conference 2012
Nov 2011	Keynote speech, Pacific Rim Symposium on Advances in Image and Video Technology (PSIVT) 2011
May 2010	Keynote speech, Korea Multi-Media Society, Cheongju, Korea
July 2006	Invited talk, Meeting on Image Recognition and Understanding (MIRU), Sendai, Japan, 2006
Feb. 2006	Invited talk, Frontiers of Computer Vision, Tokushima, Japan.
Sep. 2004	Invited talk, International Symposium on Robotics and Automation in Construction, Jeju, Korea.

ACTIVITIES

Sep 2021	Area chair, Thirty-fifth Annual Conference on Neural Information Processing Systems (NeurIPS), 2021
Oct 2019	Program chairs, IEEE International Conference on Computer Vision (ICCV), Seoul, Korea, 2019
Oct 2017	Area chair, IEEE Conference on Computer Vision (ICCV), 2017
Jun 2016 ~ Aug 2018	The Founding president, Korean Computer Vision Society
Jan 2016	President, Korean Robotics Society
Nov 2014 ~ Present	President, Asian Federation of Computer Vision
Nov 2015	General co-chairs, Pacific Rim Symposium on Advances in Image and Video Technology (PSIVT) 2015
Nov 2012	General Chair, Asian Conference on Computer Vision (ACCV) 2012
May, 2009	IEEE-ICRA2009 Senior PC Member
2008~2012	Founding Associate Editor-in-Chief, IPSJ International Journal of Computer Vision and Applications,
Jan. 2007	Program Co-chair, Asian Conference on Computer Vision (ACCV'07), Tokyo.
Jan. 2006	Area chair, Asian Conference on Computer Vision 2006.
Oct. 2005	Program Co-chair, IAPR Conference on Machine Vision Applications, Tsukuba, Japan.
2004~2014	Editorial Board, International Journal of Computer Vision.

REPRESENTATIVE PUBLICATION

Please refer to <http://rcv.kaist.ac.kr/publications.html> for details

BOOKS

- [1] Jean-Charles Bazin and In So Kweon, “Catadioptric Vision for Robotic Applications”, Lambert Academic Publishing, 2011.
- [2] Jun-Sik Kim and In So Kweon, “Metric Invariants for Camera Calibration”, Lambert Academic Publishing, 2011.
- [3] Sungho Kim and In So Kweon. Object Identification and Categorization with Visual Context, Verlag Dr. Muller, 2008.

10 Most Cited Articles

[Cbam: Convolutional block attention module](#)

S Woo, J Park, JY Lee, IS Kweon

Proceedings of the European conference on computer vision (ECCV), 3-19, 2018

[Adaptive support-weight approach for correspondence search](#)

KJ Yoon, IS Kweon

IEEE transactions on pattern analysis and machine intelligence 28 (4), 650-656, 2006

[High quality depth map upsampling for 3d-tof cameras](#)

J Park, H Kim, YW Tai, MS Brown, I Kweon

International Conference on Computer Vision, 1623-1630, 2011

[A tensor-based algorithm for high-order graph matching](#)

O Duchenne, F Bach, IS Kweon, J Ponce

IEEE transactions on pattern analysis and machine intelligence 33 (12),

[Multispectral pedestrian detection: Benchmark dataset and baseline](#)

S Hwang, J Park, N Kim, Y Choi, I So Kweon

Proceedings of the IEEE conference on computer vision and pattern

[Accurate depth map estimation from a lenslet light field camera](#)

HG Jeon, J Park, G Choe, J Park, Y Bok, YW Tai, I So Kweon

IEEE conference on computer vision and pattern recognition, 2017

[Locally adaptive support-weight approach for visual correspondence search](#)

KJ Yoon, IS Kweon

IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2005

[Pixel-level domain transfer](#)

D Yoo, N Kim, S Park, AS Paek, IS Kweon

European conference on computer vision, 517-532, 2013

[Terrain mapping for a roving planetary explorer](#)

IS Kweon, M Hebert, E Krotkov, T Kanade

IEEE International Conference on Robotics and Automation, 997-1002, 1990.

[Vpgnet: Vanishing point guided network for lane and road marking detection and recognition](#)

S Lee et al.,

Proceedings of the IEEE international conference on computer vision, 1947-1955, 2017.